

# Bearing Fault Classification for Predictive Maintenance Using Windowed Vibration Features and Random Forest

Amr Khalid Kamel Subeih  
School of Engineering  
University of the West of England  
Bristol, UK  
Amr2.Subeih@live.uwe.ac.uk

**Abstract**—Bearing faults are a leading cause of unplanned downtime in rotating machinery, driving major operational and financial losses across the industry. This paper presents a predictive maintenance solution based on a supervised machine learning pipeline for four-class bearing fault classification (Normal, Inner-race, Outer-race, Ball fault) applied to the Case Western Reserve University (CWRU) dataset. Drive-end accelerometer signals sampled at 48 kHz across four motor speeds were divided into train/validation/test sets (train: 1797 and 1772 RPM; validation: 1750 RPM; unseen test: 1730 RPM) and segmented into non-overlapping windows of 1024 samples to avoid data leakage. Seven time-domain statistical features, including RMS, kurtosis, and peak value, were extracted per window. A Decision Tree baseline achieved 97.8% validation accuracy, while the main Random Forest model reached 100% accuracy and macro-F1 on both the validation and unseen 1730 RPM test set, representing a 2.2% improvement. However, performance dropped to 56.2% accuracy and 0.56 macro-F1 on an unseen 21 mil fault diameter severity test, showing the model did not generalise to unseen fault severity. These results confirm that time-domain features provide fault signatures resilient to RPM variation, making them suitable for condition-based maintenance systems; however, further severity testing is required before any real-world deployment.

**Index Terms**—bearing faults classification, CWRU dataset, predictive maintenance, Random Forest, Decision Tree, supervised learning

## I. INTRODUCTION

Predictive maintenance of rolling-element bearings aims at reducing unplanned downtime and focuses on detecting failure-prone components by relying on raw vibrational data collected from sensors at various locations on the bearing. Bearing defects are responsible for approximately 40-50% of all electric motor failures [1], and unexpected bearing-related downtime causes multibillion-dollar losses in the manufacturing industry annually [2]. Accurate and early fault detection is of critical importance to detect faults before failure occurs, and in an industrial mechatronics context, systems such as pumps, conveyors, and automated machinery can suffer from undetected bearing degradation, which might lead to reduced performance, unexpected shutdowns, and increased maintenance costs [2]. A major engineering challenge is that machine-learning fault classifiers perform well only when the training and testing data are collected under the same operating

conditions [3]. However, in industry, this is unrealistic, as machines often operate at different rotational speeds and exhibit fault severities that the model was not trained on. A machine-learning model that cannot exhibit generalisation has limited practical value for deployment in real systems. This paper is constrained to datasets collected under controlled conditions from offline sources, rather than real-time sensor data, and assumes that the available fault labels and operation-speed metadata are accurate. The CWRU Bearing dataset [4] is a well-known benchmark for fault diagnosis studies under controlled conditions, and the CWRU-Bearing-Numpy repository used in this paper provides the dataset in a cleaned format (.npz) [15], [16]. The main task of this work is to develop a machine learning model and evaluate its performance on fault-class classification across two generalisation approaches: (i) speed generalisation, by examining whether a model trained on specific RPMs correctly predicts the class for the same fault severity at different unseen RPMs (rotational speed); and (ii) severity generalisation, by examining whether the model can still be valid if tested with unseen severity. The success of the main approach (speed) is defined by a macro-F1 threshold of 0.95, adopted as a reasonable minimum for industrial requirements.

## II. MAIN CONTRIBUTIONS

The contributions of this research paper are as follows:

1. A reproducible train/validate/test split method was constructed by separating the data by RPM, which ensures each split uses a different RPM, preventing data leakage and enabling evaluation across unseen operation speeds.
2. A seven-feature time-domain pipeline was implemented through fixed-length windowing and statistical feature extraction.
3. Decision Tree and Random Forest models were evaluated using accuracy and macro-F1 metrics on unseen data.
4. A second evaluation of severity generalisation on unseen fault diameter, which exposed a severity-dependent feature collapse as severity shifts.

### III. RELATED WORK

Bearing fault diagnosis studies have long relied on vibration analysis because faulty bearings disturb signal characteristics — such as signal energy (RMS), impulsiveness (kurtosis), and statistical time-domain indicators — that are in a form that can be measured and classified [6]. Building on this, bearing fault detection utilises both signal-processing and data-driven methods. Classical approaches depend on frequency-domain features such as envelope analysis and bearing characteristic defect frequencies extracted via Fast Fourier Transform [7]. While physically interpretable, it relies heavily on expert knowledge and struggles under operating conditions of varying statistical properties. In comparison, Support Vector Machines trained on time-domain statistical features, such as RMS, kurtosis, and peak values, have demonstrated strong performance in bearing fault classification, with studies reporting accuracy exceeding 95% on the CWRU dataset [8], demonstrating that classification based on simple statistical features can be highly effective under controlled laboratory conditions. Deep-learning architectures, most notably Convolutional Neural Networks (CNNs), applied directly to raw vibration data points acquired from sensors, have demonstrated a near-perfect accuracy on the CWRU dataset [9]. However, it requires a huge amount of computational energy, and its decision-making process is difficult to interpret. While it has shown that it partially addresses cross-generalisation problems and contributes to a reliable model performance, it does so at the cost of the complexity of the process and the difficulty of validating the model behaviour.

Among classical classifiers, a Decision Tree provide a transparent baseline structure that mimics the human decision-making process, which can be traced and easily interpreted. Random Forest improves on robustness and accuracy by training a large number of decorrelated trees while maintaining a relatively low sensitivity to hyperparameter selection [10], and their built-in feature-importance measures provide a score for each feature that enables a clear assessment of which most significantly influence the model's accuracy. This ensemble approach is particularly effective in feature spaces with nonlinear class boundaries and redundant predictors, ensuring a stable classification performance even when individual features are highly correlated [11]. Model choice in this study is therefore made by not only considering predictive performance, but also other factors that prioritise the need for an interpretable and fast decision pathway. These considerations make Decision Tree and Random Forest a compatible model for the statistical features of the datasets evaluated in this paper. Macro-averaged F1 is also used as an evaluation metric that ensures that each fault class equally contributes to the model score, so that the minority fault classes' performance is not dominated by the majority class performance under class imbalance.

Several methodological concerns that occasionally recur in different papers are based on the CWRU dataset. First, data leakage can occur when overlapping data exposes a portion of the test data to the training process, causing the model

to memorise the data rather than learn patterns. Second, the CWRU dataset was collected under controlled conditions with minimal noise and presented as a corrected and reduced set of data, which does not fully represent a realistic industrial environment [4], where background vibrations are common. Moreover, few studies explicitly distinguish cross-severity generalisation from cross-speed generalisation, and this paper addresses this gap [12]. Also, feature distribution shifts as it moves across severity, demonstrating that a model trained on a single severity may not generalize to different severities, and in an industry where safety is a critical requirement for any deployed system or model, it is essential for the user to understand fully how the decisions are made and have transparency regarding the conditions under which the model does or does not operate reliably.

### IV. METHODOLOGY

#### A. Dataset and Preprocessing

The CWRU dataset [15], [16] consisted of sampled vibration signals stored in pre-cleaned (.npz) files. Each file contained up to three accelerometer channels: Drive-End (DE), Fan-End (FE), and Base (BA), which were recorded from different sensor locations and stored as a (float64) time-series array. The drive-end (DE) signal was selected as the primary channel for analysis because it was the most consistently available across the dataset files and aligns with its common use as the standard benchmark in CWRU-based classification studies. Data were grouped into four labels that represent the health status of the bearing: Normal, Inner Race (IR), Outer Race (OR@3), and Ball Fault (B). All signals were sampled at 48kHz, providing fine temporal resolution of the recorded vibration signals.

These labels were distributed across multiple files recorded at different motor speeds (1797, 1772, 1750, and 1730 RPM) and fault severities of (7, 14, 21, and 28 mil). The file mapping covered the class structure that defined the model's prediction targets, using a strictly defined data split: 1797 and 1772 for training, 1750 for validation, and 1730 for testing on the main model, all of which fall under the same severity level (7mil). A second independent test was conducted at 21 mil fault diameter and 1730 RPM to evaluate the severity generalisation. The model was evaluated without retraining, and the same fitted scaler was applied, ensuring that it was tested on fault-severity data not seen during training.

No additional data cleaning was required, as it is stated that the CWRU dataset contains pre-processed and validated vibration signals without missing or corrupted samples [15]. The DE signals were segmented into 1024 non-overlapping windows, consistent with common practice in CWRU-based studies [4]. Table I shows the number of windows allocated to each data split. The windows were divided into train/validate/test sets, with each split assigned by RPM conditions. Because windows extracted from the same RPM-file are highly correlated, this split strategy focused on preventing the model from memorising the data patterns, and focused on

learning them. This ensures that no test window comes from the same data file as any training window.

Seven time-domain statistical features were computed per window: mean, standard deviation, variance, RMS, peak value, skewness, and kurtosis. The statistical features extracted from each window served as the input to the supervised learning model, as they were considered a well-established and computationally lightweight representation for this type of classification. Feature scaling was also applied by fitting the scaler only on the training set, and then applying it on the validation and test sets, as this was considered an appropriate and common leakage-control step that prevents the mean and variance of unseen data from leaking into the training process, and influences the model's behaviour.

TABLE I  
DATASET PARTITION SUMMARY

| Split    | RPM(s)     | Diameter | Windows | Role                    |
|----------|------------|----------|---------|-------------------------|
| Train    | 1797, 1772 | 7 mil    | 2730    | Model fitting           |
| Validate | 1750       | 7 mil    | 1896    | Model selection         |
| Test (1) | 1730       | 7 mil    | 1899    | Speed generalisation    |
| Test (2) | 1730       | 21 mil   | 1899    | Severity generalisation |

### B. Models and Baseline

Two supervised classification approaches were implemented to identify and classify bearing faults. A Decision Tree as the baseline and a Random Forest as the main model were chosen based on the fact that both models belong to the same tree-based family, demonstrating a controlled comparison in which any performance differences mainly reflect the multi-tree structure rather than differences in features or evaluation.

**Baseline — Decision Tree (DT):** A Decision Tree using default hyperparameters (Gini impurity, no depth constraint, uniform class weights, `random_state = 42`) was used as a reference model for comparative analysis. These default configurations were kept to ensure a simple and transparent model. This approach reflects an industrial practice in which simple rule-based models are designated for classification, as they provide a clear and traceable sequence of value-based splits, which is an important factor in safety-related maintenance settings [13].

**Main Model — Random Forest (RF):** The random forest model (`n_estimators=300`, `max_features='sqrt'`, `class_weight='balanced'`, `n_jobs=-1`, and `random_state=42`) was used with these tuned hyperparameters. Conceptually, this model is represented as the extension of the decision tree approach into a more complex and robust tree structure of multiple aggregated trees. Each tree was trained on a randomly resampled subset of the original training data (bootstrap sample), achieved by randomly sampling windows with replacement. At each split, the algorithm checks only a random subset of the square root of  $p$  features ( $p = 7$  in this paper) [10]. These two causes of randomness in the

model's training process resulted in a reduced correlation between trees and produced a more diverse aggregated structure, which is a primary factor in improving the generalisation of the model. Combining the predictions of 300 trees through majority voting demonstrated a substantially reduced variance compared to a single decision tree, and the decision of setting `n_estimators=300` was a typical benchmark for models with relatively small numbers of features, in which this number of trees is a suitable value for a good-performing model. The `class_weight='balanced'` compensated for the class imbalance in the training data and prevented the model from taking biased decisions in favour of the majority classes.

Both models were trained on identical inputs (the same seven time-domain features, the same RPM-based splits, and the same validation and test sets), and this choice of approach demonstrated that this comparison isolates the influence of bootstrap aggregation and random feature selection, since these are the core structural differences between a Random Forest and a Decision Tree, ensuring the baseline comparison is fair and logical.

### C. Evaluation Design

Two independent generalisations were evaluated. The primary evaluation was a speed generalisation using Test (1), which assessed model performance on unseen 1730 RPM, and 7 mil fault diameter. A second evaluation demonstrated severity generalisation using Test (2), performed with the same 1730 RPM but with unseen severity of 21 mil fault diameter. It is important to note that speed generalisation is the primary focus of this model, but an additional evaluation was introduced to reveal the design's limitations. Performance was measured using accuracy and macro-averaged F1-score, with the main metric being macro-averaged F1-score (macro-F1) because it is appropriate for multi-class models with mild imbalance [14].

## V. RESULTS

### A. Feature Analysis

Fig. 1 presents box plots of RMS and kurtosis against all four bearing fault classes. RMS shows a clear monotonic separation between fault classes (medians): Normal ( $\approx 0.07$  g), Inner Race ( $\approx 0.29$  g), Outer Race ( $\approx 0.45$  g), and Ball Fault ( $\approx 0.15$  g). The normal class results exhibit the lowest and tightest RMS distribution, which aligns with the expectations that a healthy bearing under a consistent load will result in minimal vibration amplitudes. Through observations based on the box plot, the Outer Race fault displayed the highest median value among all classes, and this is physically consistent, as the outer race does not rotate and stays in the same position under load. As a result, every rolling element passes over the same damaged spot on each rotation, producing high-energy impacts that continuously elevate vibration levels and result in higher RMS. Inner faults lean toward an intermediate RMS, while ball faults, in comparison to others, sit at low levels of RMS due to a substantially smaller contact area and time under load. The inner race distribution displays many high

RMS outliers, presumably because the defect rotates through the loaded areas, which will generate strong impacts that periodically create sharp RMS values of up to ( $\approx 0.65$  g). These spikes inflate some of the windows, while the remaining windows remain unchanged at lower RMS values [6].

In contrast, the kurtosis boxplot captures a different but complementary bearing behaviour. The Inner Race fault has the highest median kurtosis of ( $\approx 4.0$ ), and the longest spread of values, with outliers exceeding (6.0), which is an indication of a consistent pattern of the high-impulsivity nature for the inner race defects. Although the Normal fault class shows a median near-zero on the kurtosis box plot and low variance, this supports that the behaviour observed across the bearing fault classes is consistent across features and aligns with the expected physical characteristics [6]. Outer Race faults have a kurtosis of ( $\approx 1.0$ ), clearly lower than the inner race, although it has a higher RMS. This reflects the nature of outer race impacts being fixed in the load zone and results in more steady, less impulsive vibrations than the sharp spikes of the inner race [6].

Although RMS and kurtosis, when interpreted, each have displayed a consistent physical behaviour across the bearing classes, they are clearly different from each other and capture different aspects of the signal. The box plots show that the classes are well separated, and these features have shown that they are good at distinguishing between the fault types, as classes do not overlap much, and this is an advantage for any classification work. Because the features demonstrated clear class separation at 7 mil fault severity, this provided strong justification for using them in speed generalisation analysis. Overall, the plot shows that no single feature is sufficient to identify and classify all four fault classes and a minimum of two independent feature types (energy captured by RMS) and impulsiveness (captured by kurtosis) are a requirement for reliable results. This motivated the use of seven features per window to ensure a wider coverage of statistical characteristics captured across the classes. RMS helped primarily distinguish the outer race from other classes, while kurtosis distinguished the inner race, and the ball fault is more clearly separated from the normal class through RMS than through kurtosis.

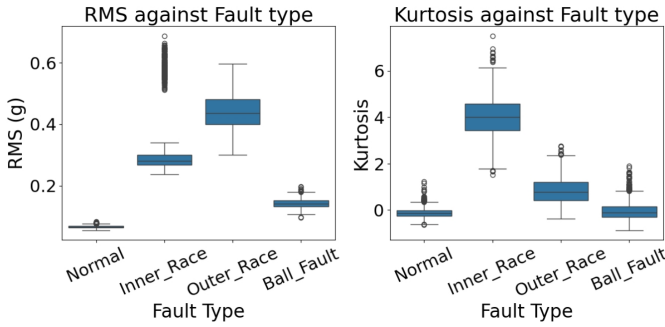


Fig. 1. Box plots of RMS and kurtosis per fault class.

The final seven features used are shown in Fig.2, which is a correlation heatmap matrix. It illustrates that std\_deviation,

variance, RMS, and peak\_value are highly correlated with each other, while skewness and kurtosis show low levels of correlation with these features and capture a more independent type of behaviour. This further supports the inclusion of the additional features, as they provide a more distinctive set of information that strengthens the model's ability to classify the bearing faults.

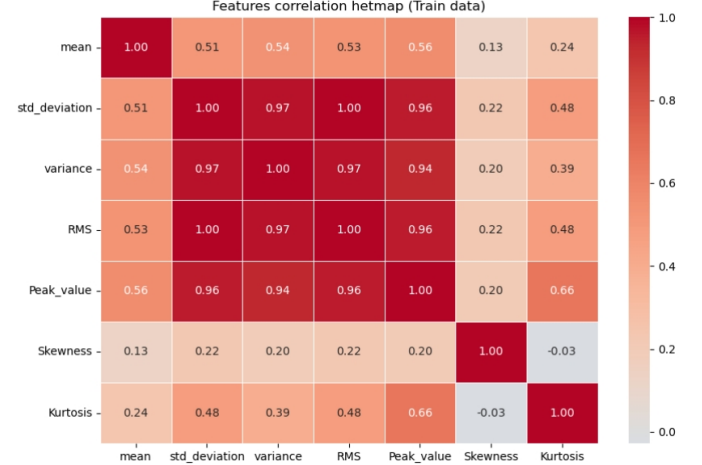


Fig. 2. Correlation heatmap of the seven extracted time-domain features.

## B. Model Performance

The Decision Tree baseline achieved a 97.8% accuracy and 0.978 macro-F1 on the 1750 validation set, as shown in (Table II). Ball Fault and Normal were classified perfectly with a score of 1.00. Inner Race recall was 0.92, and Outer Race precision was 0.92, indicating where the model's highest uncertainty lies. The inner race and outer race are the farthest from the perfect and high scores, indicating that the inner race's lower recall score was due to some windows lacking the unique characteristics of a faulty bearing, which caused the model to misclassify. On the other hand, the outer race shows reduced precision, and this can be interpreted as the presence of false positives, which is a result of other high-energy windows from other faults being mislabeled as the outer race. This highlights the limitations of a single tree model when features overlap.

TABLE II  
DECISION TREE BASELINE — VALIDATION SET (1750 RPM, 7 MIL)

| Class                              | Precision | Recall | F1   | Support |
|------------------------------------|-----------|--------|------|---------|
| Ball Fault                         | 1.00      | 1.00   | 1.00 | 475     |
| Inner Race                         | 0.99      | 0.92   | 0.96 | 474     |
| Normal                             | 1.00      | 1.00   | 1.00 | 473     |
| Outer Race                         | 0.93      | 0.99   | 0.96 | 474     |
| Macro Avg                          | 0.98      | 0.98   | 0.98 | 1896    |
| Accuracy: 0.978    Macro-F1: 0.978 |           |        |      |         |

The Random Forest (main model) has achieved 100% accuracy and 1.0 macro-F1 on both the 1750 RPM validation set and the unseen 1730 RPM Test(1) set, as shown in Table III. A (+2.2) percentage improvement was observed in the random forest model performance over the decision tree baseline. All of the classes have shown a precision, recall, and F1 scores of 1.0 across all 1896 validation and 1899 test windows. This score results from the unseen 1730 RPM test, which far exceeds the industrial threshold of 0.95 defined earlier in the paper, indicating that the model meets the deployment requirement for speed generalisation. The observed perfect classification performance of the model is consistent with the robust inter-class feature separation shown in Fig. 1, and the controlled nature of the CWRU dataset with a low level of noise, which may not capture precisely real industrial conditions and behaviours.

The consistency between the outcomes of the validation (1750 RPM) and the test (1730 RPM), where both achieve 1.0 macro-F1, confirmed that the perfect results were not caused by overfitting to the validation or test RPMs, but on the contrary it proved the model reliability, as identical performance was reproduced on a completely unseen set of data (Test(1)).

TABLE III  
RANDOM FOREST — SPEED GENERALISATION RESULTS

| Class                                                        | Prec. | Recall | F1   | Supp. |
|--------------------------------------------------------------|-------|--------|------|-------|
| <i>Validation (1750 RPM, 7 mil) Acc. 1.000 Windows: 1896</i> |       |        |      |       |
| Ball Fault                                                   | 1.00  | 1.00   | 1.00 | 475   |
| Inner Race                                                   | 1.00  | 1.00   | 1.00 | 474   |
| Normal                                                       | 1.00  | 1.00   | 1.00 | 473   |
| Outer Race                                                   | 1.00  | 1.00   | 1.00 | 474   |
| <i>Test (1) (1730 RPM, 7 mil) Acc. 1.000 Windows: 1899</i>   |       |        |      |       |
| Ball Fault                                                   | 1.00  | 1.00   | 1.00 | 477   |
| Inner Race                                                   | 1.00  | 1.00   | 1.00 | 474   |
| Normal                                                       | 1.00  | 1.00   | 1.00 | 474   |
| Outer Race                                                   | 1.00  | 1.00   | 1.00 | 474   |

In Fig. 3, the confusion matrix for both the validation at 1750 RPM and the unseen test(1) at 1730 RPM is presented, showing that all of the 1896 validation and 1899 test windows are correctly classified with no errors observed in the matrix, which reflects that the 1.0 macro-F1 results genuinely represent a perfect classification outcome for every class, without any hidden errors in specific classes. This matrix also shows that the model has learned to genuinely generalise across fault classes, rather than memorising the training data, as it maintains a stable model performance across the RPM shift from 1750 to 1730 RPM.

The near-equal class (Supp.) across all classes (473 to 477 in validation, 474 to 477 in test (1)) indicate that the perfect macro-F1 score was not a result of class imbalance, as each

class had contributed approximately equally to the metric. Furthermore, the Random Forest resolved the weaknesses observed in the Decision Tree baseline, which identified as the Inner Race recall of 0.92 and the Outer Race precision of 0.93, with both improved to reach a score of 1.0 in the main model. This shows that the feature sampling strategy produced a more robust model capable of separating the overlapping of windows between these classes.

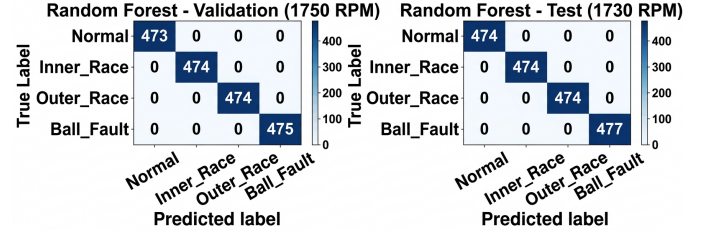


Fig. 3. Confusion matrices for the Random Forest model on both the validation and unseen Test (1).

Table IV presents results on Test (2): the same specifications applied to it, starting from the training model to applying a scaler on the 21 mil fault diameter data at 1730 RPM. Overall, it has been observed that accuracy sustained a large drop to 56.2%, which resulted in a degradation from the speed generalisation results, with classes showing non-uniform behaviour. Normal class retained a perfect F1 of 1.0, showing consistency across cross severities, while Ball Fault had an F1 value of 0.86, and a recall of 0.75. This means 25% of ball fault windows were misclassified. Furthermore, Outer Race dropped to an F1 of 0.37 and low (prec.) of 0.30, which implies that windows belonging to other classes were being predicted and misclassified as outer race. And Inner Race suffered the biggest drop, where the F1 reached (0.02) a near-zero score that indicates that the model failed to identify any inner race faults at 21 mil severity.

TABLE IV  
RANDOM FOREST — SEVERITY GENERALISATION (1730 RPM, 21 MIL)

| Class      | Prec. | Recall | F1   | Supp. |
|------------|-------|--------|------|-------|
| Ball Fault | 1.00  | 0.75   | 0.86 | 475   |
| Inner Race | 0.02  | 0.01   | 0.02 | 477   |
| Normal     | 1.00  | 1.00   | 1.00 | 474   |
| Outer Race | 0.30  | 0.49   | 0.37 | 473   |
| Macro Avg  | 0.58  | 0.56   | 0.56 | 1899  |

Accuracy: 0.562 Total Windows: 1899

This drop in performance reflects a severity-driven shift in the feature values. At 7mil, each fault class was located across a range of values that presented a distinct region in the seven-feature space (Fig.1), while at 21mil, the larger defect size produced much stronger vibrations, shifting the feature values to a range beyond the learned pattern during training. This is clearly seen in the Normal class, in which

it maintained a perfect macro-F1 of 1.0 on both the 7mil and 21mil severities, and that is an indication that the model can correctly distinguish a healthy bearing even at unseen severities, but cannot categorise the faulty class types. Overall, the results show that the model does not generalise to unseen fault severities, and the outcome performance of 0.56 macro-F1 falls much below the industrial threshold of 0.95. This implies that the model is valid for speed generalisation at constant severity but can not be deployed in cross-severity industrial environments without further retraining.

TABLE V  
HEADLINE COMPARISON ACROSS ALL EVALUATION CONDITIONS

| Condition             | Diameter | Accuracy | Macro-F1 |
|-----------------------|----------|----------|----------|
| DT Baseline (val)     | 7 mil    | 0.978    | 0.978    |
| RF Validation         | 7 mil    | 1.000    | 1.000    |
| RF Test(1) — Speed    | 7 mil    | 1.000    | 1.000    |
| RF Test(2) — Severity | 21 mil   | 0.562    | 0.560    |

## VI. DISCUSSION

### A. Performance Interpretation

The perfect performance on the unseen 1730 RPM Test (1) can be interpreted as a genuine model generalisation across relatively small RPM variations. The seven time-domain features have captured fault-related amplitudes and impulsiveness behaviours consistently, and based on the observed outcomes, the features remained stable as the shaft speed shifted by relatively small margins, and therefore, it can be assumed that this amount of RPM shift from training speeds (1797, 1772) to testing speed of (1730 RPM) did not alter the underlying vibration patterns of the various class faults, contributing to a model that can recognise the same learned pattern during testing.

In addition, the Random Forest further improves the stability of the outcome results. By aggregating 300 trees, the model has learned to accommodate variations in features across several RPMs, enabling it to capture the underlying fault characteristics rather than simply memorising patterns.

The severity generalisation Test (2) result of 56.2% accuracy is not considered a model failure in this paper, but rather an expected and informative result that highlighted the severity dependence of time-domain features. The main scope of this paper was focused on speed generalisation, while the introduced additional Test (2) serves as a diagnostic evaluation to test the model limitations. The results in Tables III and IV were a clear indication of how the model performed under seen and unseen severities: 1. Normal achieved a (1.0 F1) score across both tests because healthy bearings do not produce features that display fault characteristics; 2. Ball Fault show some moderate level of generalisation (0.86 F1) on unseen severity, which suggests that its behaviour and pattern are consistent to a certain degree moving from a 7 mil to a 21 mil fault diameter; 3. Inner Race collapses to 0.02 F1 and

Outer Race to 0.37 F1, where this can be concluded as a result of feature vectors being shifted to a region that was unseen during training. This shift aligns with documented findings [5] that models trained at fixed severities fail to generalise across fault diameters.

These results are consistent with previous studies and findings [6], which stated that RMS and kurtosis shift substantially across fault severities, especially in Inner Race. This implies that the model requires further training in areas where multiple severities are expected to avoid any uncertainties in the model performance if deployed. Also, they can be interpreted as a validation that the model was performing genuine classification and reflects a real generalisation based on learned patterns, not any overfitting or memorisation of training data. This was based on the variation in the degraded performance of Test (2), especially in each specific class, which can be interpreted in relation to actual physical properties.

### B. Limitations

The CWRU dataset presents data collected under controlled laboratory conditions, with no background vibrations or typical external interference in real industrial settings. As the dataset contains minimal noise, the model's robustness to real industrial scenarios remains unverified. All the data was pre-recorded offline, so there is no real-time supply and tracking of the captured signals. Moreover, the train RPM range is narrow, with 2730 windows drawn from only two RPM conditions, whereas a larger RPM range may have a large effect on model performance. The whole process builds on the assumption that the provided fault labels and severity metadata are accurate.

Fig. 4 presents the Random Forest feature importance bar chart, ranking the features by the Mean Decrease in Impurity (MDI). It shows that the skewness contribution is of very little importance to the model performance, followed by the mean, which is ranked 6th, with both features contributing well below the remaining five features (RMS, standard deviation, variance, peak value, and kurtosis) that the model is primarily driven by. The correlation heatmap (Fig. 2) has revealed that standard deviation, variance, RMS, and peak values are strongly correlated, which could indicate the presence of redundant features. While seven features were selected for the model to be able to capture a wider coverage of statistical features, no formal feature selection analysis was conducted to identify the minimal sufficient set of features, and this is presented as a gap in the current paper.

Most importantly, the model has demonstrated no severity generalisation, with the performance results dropping from a macro-F1 of 1.0 to 0.56 on the 21 mil unseen fault diameter. This is a clear indication that the learned features are severity-dependent and would require further retraining on various severities for the model to be deployed across different fault diameters. Severity dependence serves as the most critical limitation in the model, as it directly influences the range of fault conditions the model can reliably classify.



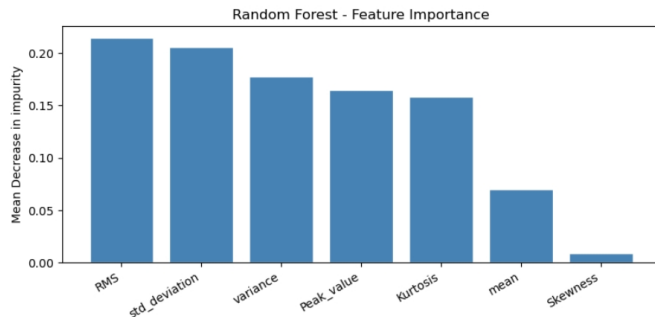


Fig. 4. Random Forest feature importance ranked by Mean Decrease in Impurity (MDI).

### C. Ethics, Safety, and Responsible Engineering

In industrial settings, bearing performance and maintenance are critical to the operation of any rotary machine. Work done through bearing fault classification models has shown that the risk of false negatives can lead to a broad range of failures and injury in some cases, while false positives can trigger significant operational inefficiencies and wasted maintenance efforts [13]. The applicability of this model in industrial scenarios of bearings at high severity levels poses an operational risk, as the model may misclassify the Inner Race faults as Normal, and this can result in dangerous false negatives. Therefore, deployments of the model must take into account the expected range of fault severities for any potential bearings to be tested.

### D. Engineering Implications

The model design has a low computational footprint (lightweight to compute and fast), which displays its potential for real-time deployment. However, test (2) results highlighted the practical deployment constraint that the model has, and to construct a successful maintenance plan, these gaps need to be managed by cost-sensitive regulations or algorithms that can balance between the severity risk and the maintenance cost for an effective and successful deployment in industry [13].

## VII. CONCLUSION AND FUTURE WORK

This paper presented a supervised bearing fault classification model that was assessed based on two generalisation approaches using the CWRU benchmark dataset. Key findings and future work are presented below.

- **Speed generalisation achieved:** The Random Forest has achieved a 100% accuracy and 1.0 macro-F1 on both the 1750 RPM validation data and the unseen 1730 RPM Test (1). It outperformed the Decision Tree baseline with scores of 97.8% accuracy and 0.978 macro-F1, which displayed a reliable model against RPM-shifted classification problems under a fixed severity of 7mil fault diameter.
- **Severity generalisation:** The secondary test (Test (2)) was conducted to identify the boundaries of the model. The unseen 21 mil severity was used in the test, and

results displayed a drop in accuracy to 56.2 %. While Normal generalised perfectly at 1.0 F1, Inner Race and Outer Race dropped sharply (0.02 and 0.37 F1), showing how the change in fault severities changed the feature values and therefore, the model performance.

- **Engineering significance:** The use of time domain features has supported a resilient speed generalisation methodology within a specific RPM range, but under a fixed severity value, which does pose some restrictions on the deployability of the model.
- **Future data:** Train the model for more iterative trials on different fault diameters (7, 14, 21, 28 mil) to expand the classification capacity of the model.
- **Future methods:** Investigate the feature selection and its dimensionality to identify the minimal sufficient set of features and explore frequency domains to identify a path for a more reliable and improved cross-severity model.
- **Future deployment:** Evaluate the model against a different and independent bearing dataset to determine if the characteristics and patterns of fault bearings that have been learned through the features from the CWRU dataset can work with different configurations, bearing geometries, and RPM ranges.

### Reproducibility

The entire workflow and implementation, including data loading, feature extraction, model training, and model evaluation, are included in the Jupyter Notebook artefact. Required libraries: (scikit-learn, Numpy, pandas, SciPy, Matplotlib, and Seaborn), which run on Python 3.13.5. All figures and tables were generated within the notebook, and their results are presented in the appropriate format in the paper. CWRU bearing dataset files (.npz) must be downloaded from [https://github.com/srigas/CWRU\_Bearing\_NumPy] if needed, but the files required for this paper are in the same zip folder. Run instructions are provided in the README.md file.

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